

GROUP O

Feasibility Study and Risk Register

Multi-Site Smart Soil and Environmental Monitoring System

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1. Purpose and Method

This document records whether the Multi-Site Smart Soil and Environmental Monitoring System can realistically be delivered with the time, skills and resources available, and what could prevent that from happening.

Six areas are assessed: technical, operational, economic, schedule, legal and ethical. Each reaches one of three conclusions — feasible, feasible with changes, or not currently feasible — and each states the change required where one applies. Twelve risks are then recorded, each rated for likelihood and impact, and each given a mitigation, a contingency and a named owner.

The assessment draws on the stakeholder requirements document of 20 August 2026, the written stakeholder responses of 27 August 2026, and the consultation meeting of 19 August 2026.

2. Technical Feasibility

The application layer uses frameworks the delivery team has worked with in prior coursework. The required functions are conventional application development: authenticated access, list and detail screens, form-driven management, filtered queries, charting and export.

The uncertainty that existed at the start of the project concerned retrieving data from a system operated by a separate party. The stakeholder confirmed in writing on 27 August 2026 that live integration is not required for this version and that representative sample data is acceptable. That removed the dependency the delivery team did not control.

Skills still to be developed are database access rules, scheduled backend functions, and charting in both frameworks. Time for this is allocated before development begins.

Question	Finding
Does the team have the required Flutter skills?	Yes. Covered in prior coursework. The required screens are conventional list, detail and chart views.
Does the team have the required ASP.NET skills?	Yes. Covered in prior coursework. The required functions are form-driven management screens, filtered queries and report export.
Can the team work with Supabase?	Yes for database, authentication and queries. Row-level access rules and scheduled functions are new and require learning time.
Can both applications use the same backend?	Yes. Both connect to one Supabase project through its client libraries and share the same tables and authentication.
Are development machines available?	Yes. Every member has a machine capable of running the Flutter and .NET toolchains.
Are test devices available?	Yes. At least one physical Android device is available for the demonstration.
Is reliable internet access available?	Yes, for the team. Both applications and the backend require a connection.
Which skills still need to be learned?	Scheduled backend functions, row-level security policies, charting in both frameworks, and export to CSV and PDF.
Which functions may be too difficult?	None. The technical uncertainty was dependency rather than difficulty, and that dependency has been removed for this version.

Decision	Feasible
Reason	The required technologies are within the team's demonstrated capability, and the external dependency that carried the technical risk has been removed from this version.

3. Operational Feasibility

The system requires no change to how measurements are produced and no manual data entry by users. Users read information rather than supplying it, which keeps the burden of adoption low.

Question	Finding
Will the prototype address the stakeholder's problem?	Yes. The stated problem is delayed identification of change across three locations. The system brings the readings together, compares them against configured ranges and reports what needs attention.
Who will use the mobile application?	Agricultural staff and field workers at the sites, and the site or farm manager.
Who will use the web application?	Administrative users managing configuration, and researchers analysing conditions across the sites.
Do the intended users have suitable devices?	Yes. Mobile users have Android devices; administrative users work at computers.
Do users have suitable internet access?	Yes at the office. Connectivity at the sites has not been separately confirmed, which is why offline capability is recorded as a later enhancement rather than a requirement.
Will the solution fit current activities?	Yes. No change is required to how sensor data is produced, and users consume information rather than entering it.
Will users require training?	Minimal for the mobile application, which is read-only. Brief guidance is required for threshold configuration and user management on the web application.
Are any functions unsuitable for the intended users?	Threshold configuration must be restricted to administrators. If available to all users, one person changing a value would alter the alerts everyone else sees.

Decision	Feasible
Reason	The solution fits the organisation's existing activities, the two user roles were confirmed by the stakeholder, and the functions map directly to the difficulties described.

4. Economic Feasibility

This assessment covers whether the delivery team has access to the resources required to complete the prototype. A full business financial analysis is not required for this project.

Resource	Available	Position
Internet access or data	Yes	Available to all members through campus and personal connections
Transport to the stakeholder	Yes	The stakeholder is based on campus; engagement has been by meeting and email

Resource	Available	Position
Development computers	Yes	Every member has a machine capable of running both toolchains
Test devices	Yes	At least one physical Android device for the demonstration
Required software	Yes	Flutter, the .NET SDK, Visual Studio Code and Git are free. Microsoft Project is provided through the institution.
Supabase resources	Yes	Free tier. Database size is monitored as readings accumulate — see risk R11.
Presentation resources	Yes	Venue, display and connection available for the demonstration
Sensor hardware	Not required	The equipment is owned and operated by a separate party. No procurement falls to the delivery team.

Decision	Feasible
Reason	All required software operates within free or institutionally provided licences, no hardware procurement falls to the delivery team, and every resource needed is available.

5. Schedule Feasibility

Approximately three weeks are available for development across a mobile application, a web application and a shared backend. The delivery team has eight members, which permits parallel work, but components developed separately require time to combine and that time is planned rather than assumed.

Removing the live integration from this version reduces the work by one component and removes the item that could not be designed until an external data format was known.

Question	Finding
Is there time for system design?	Yes. Two weeks are allocated to architecture, UML, database design and interface prototyping in Phase 2.
Is there time for Flutter development?	Yes, provided the backend and database schema are complete before mobile development begins.
Is there time for ASP.NET development?	Yes, on the same condition. The web application carries more screens and should start no later.
Is there time for Supabase integration?	Only if the schema is finalised first. Both applications depend on it, so it is the most schedule-critical item.
Is there time for testing and correction?	One week is allocated. Sufficient only if integration is tested continuously rather than at the end.
Is there time for stakeholder feedback?	Yes at the Phase 2 design review, and again at the prototype evaluation in Phase 3.
Are too many functions included?	The additional features proposed by the team exceeded what three development weeks can absorb. These have been simplified, postponed or removed — see the refined scope in the specification.

Required changes

1. Development is planned backwards from the fourteen acceptance criteria. Work not contributing to one of them is deferred.
2. The database and access rules are completed before parallel application development begins, because both applications depend on them.
3. Time for combining components is reserved within the development window rather than added at the end.
4. Each of the mobile, web, backend and provisioning components has a named owner.

Decision	Feasible with changes
Reason	The available time is sufficient only if scope is restricted to the acceptance criteria and the shared backend is completed before parallel development begins. Without both changes the schedule is not achievable.

6. Legal Feasibility

Question	Finding
Will the prototype collect personal information?	Only user account details — email address and assigned role. Sensor measurements are not personal information.
Is all the information necessary?	Yes. Account details are required for authentication and role-based access. No additional personal information is collected.
Who may access the information?	Access is restricted by assigned role. Administrators may change configuration; standard users may read only.
Will test data be used during development?	Yes. Only test accounts and representative sample data are used in the prototype and in the demonstration.
How are credentials protected?	Held in environment variables outside the source repository. A .gitignore file was committed before any configuration file was created. No key is embedded in the client applications.
Does the project involve confidential information?	Site details and operational data belong to the stakeholder organisation. Only what the stakeholder has supplied in writing is reproduced in project documentation.
Is private information kept out of the repository?	Yes. No password, API key, service role key, telephone number or identity number has been committed at any point.

Handling of personal information is subject to the Protection of Personal Information Act. Restricting the prototype to test accounts keeps the project outside the handling of real personal information.

Decision	Feasible
Reason	The prototype handles no real personal information, does not access third-party data in this version, and credential protection is in place.

7. Ethical Feasibility

Question	Finding
Does the stakeholder understand the purpose?	Yes. The stakeholder understands that this is a prototype delivered within a defined period, not a production system with ongoing support.
Is unnecessary personal information collected?	No. Only the account details required for authentication and role assignment.
Could the solution negatively affect users?	The identified risk is over-reliance. If the system presented threshold comparison as agronomic advice, a user might act on a recommendation the system is not qualified to make. The system therefore reports measured values against configured ranges and offers no crop or treatment recommendation.
Is information used only for its purpose?	Yes. Measurements are used to display conditions, raise alerts and produce reports for the organisation, and for no other purpose.

Question	Finding
Does the prototype reflect stakeholder needs?	Yes. The stakeholder confirmed the findings and approved the requirements on 3 September 2026.
Are users treated fairly?	Access is assigned by role rather than by individual. Incomplete functions are documented rather than presented as working.

Decision	Feasible
Reason	No unnecessary information is collected, information is used only for its stated purpose, and an explicit boundary is recorded against advice the system is not qualified to give.

8. Feasibility Summary

Area	Decision	Main reason	Change required
Technical	Feasible	Required technologies within the team's capability; external dependency removed from this version	None
Operational	Feasible	Users read information rather than entering it; two roles confirmed by the stakeholder	Configuration restricted to administrators
Economic	Feasible	Software free or institutionally provided; no hardware procurement	None
Schedule	Feasible with changes	Approximately three development weeks across three components	Restrict scope to the acceptance criteria; complete the backend before parallel work
Legal	Feasible	No real personal information handled; no third-party data accessed in this version	None
Ethical	Feasible	Explicit boundary recorded against agronomic advice	None

9. Feasibility Conclusion

The proposed project is feasible with changes.

The main strengths of the project are that the required technologies are within the delivery team's demonstrated capability, that the software and services needed are available at no cost, that no hardware procurement or maintenance falls to the delivery team, and that the system requires no change to how the organisation currently produces its measurements. Users read information rather than supplying it, which keeps the burden of adoption low.

The main limitation identified is time. Approximately three weeks are available for development across a mobile application, a web application and a shared backend, and eight people working in parallel carry an integration cost that must be planned rather than assumed.

To ensure that the project can be completed within the available period, the delivery team will plan development backwards from the fourteen acceptance criteria and defer work that does not contribute to one of them, complete the database and access rules before any parallel application development begins, reserve time for combining components within the development window, assign a named owner to each component, and defer the additional features recorded in the refined scope until the required functions are complete.

10. Risk Rating Scale

Rating	Likelihood	Impact
1	Unlikely to occur	Minor delay or inconvenience
2	Possible	Noticeable effect requiring action
3	Likely	Serious effect on delivery or quality

Risk score = likelihood × impact. Scores of 1 to 3 are Low, 4 to 6 are Medium, and 7 to 9 are High.

A risk is something that may happen and would affect the project if it did. Something that has already happened is not a risk but an issue, and is recorded in the Challenges and Actions sheet of the project progress tracker.

11. Project Risk Register

ID	Risk	Category	L	I	Score	Level
R01	Development time may be insufficient for the full scope, leaving required functions incomplete	Schedule	3	3	9	High
R02	Components developed in parallel may fail to integrate, so that parts work individually but not together	Technical	2	3	6	Medium
R03	Credentials may be committed to the repository, exposing keys that deletion alone cannot withdraw	Security	2	3	6	Medium
R04	Threshold values may prove unsuitable for the sites, producing alerts that do not indicate a real problem	Quality	2	3	6	Medium
R05	Contribution may be uneven across an eight-member team, concentrating work on a few people	Team	2	2	4	Medium
R06	Scope may expand during development as features are added mid-build	Scope	2	2	4	Medium
R07	A team member may become unavailable, leaving a component without an owner	Team	2	2	4	Medium
R08	The stakeholder may be unavailable when design feedback or evaluation is required	Stakeholder	2	2	4	Medium
R09	Sample data may not resemble real readings closely enough to demonstrate the system convincingly	Quality	2	2	4	Medium
R10	Learning time for access rules and scheduled functions may exceed the allowance made	Technical	2	2	4	Medium

ID	Risk	Category	L	I	Score	Level
R11	Free service limits may be reached as readings accumulate	Resource	1	2	2	Low
R12	Later live integration may reveal a data structure the current design cannot accommodate	Technical	2	1	2	Low

12. Risk Response and Ownership

ID	Mitigation — before it happens	Contingency — if it happens	Owner
R01	Plan development backwards from the acceptance criteria; complete the backend before parallel work; reserve integration time explicitly	Deliver Must Have requirements only; document incomplete functions rather than presenting partial features as complete	Caitlin Freeman
R02	Finalise the database structure before application development begins; agree field names and types in writing; integrate early rather than at the end	Reserve time in the final week for correction; prioritise the integrated demonstration over individual features	Nido Maphosa
R03	Commit an ignore file before any configuration exists; hold keys in environment variables; brief every member before their first commit	Rotate the exposed key immediately. The file must be removed and the credential replaced, as deletion does not remove it from history	Nido Maphosa
R04	Record the basis of every threshold; make all values configurable per location; state the known limitations openly	Adjust the values with the stakeholder and record the change in the requirements change log	Machedi Mamello
R05	Assign named responsibility for every task; record contributions with evidence; review progress at a fixed weekly meeting	Reallocate outstanding work and record the reallocation in the progress tracker	Caitlin Freeman
R06	Record every requirement with a priority; treat Should Have and Could Have as excluded from the initial build; log any change with a reason	Return added work to the backlog and document the decision	Ayatolla Pikwa
R07	Ensure at least two members understand each component; keep all work in the shared repository rather than on individual machines	Redistribute the affected work and record the change	Caitlin Freeman
R08	Agree design feedback and evaluation dates in advance; put every request in writing	Proceed on previously confirmed findings and request written feedback	Caitlin Freeman
R09	Generate values within realistic ranges, including occasional threshold breaches and periods of sensor inactivity	Adjust the generator and demonstrate against a prepared data set	Thobani Mjiyakho
R10	Allocate explicit learning time before development begins; build a small working example of each first	Simplify the affected function and document the simplification	Mbasa Tyam
R11	Monitor database size; store readings at the defined interval rather than at maximum frequency; use summarised data for long-range charts	Archive older readings and reduce the provisioning interval	Mbasa Tyam
R12	Isolate all provisioning in one component so that a replacement	Adapt the replacement component only; no other	Nido Maphosa

ID	Mitigation — before it happens	Contingency — if it happens	Owner
	affects nothing else	component changes	

13. Highest Risks

R01 is the only High risk. R02, R03 and R04 share a score of 6 and are treated as the next priority. R04 is singled out below because alerts are the purpose of the system and the thresholds decide whether they can be trusted.

Rank	Risk	Score	Why it matters	What the team is doing now
1	R01 — Development time is insufficient	9	Approximately three weeks are available for fourteen acceptance criteria across three components. Scope control is the only available response and it must be applied before development starts, not during it.	The refined scope is treated as closed. The backend is sequenced first. Each component has a named owner.
2	R02 — Components fail to integrate	6	The prototype is evaluated as one integrated system. Components that work individually but do not combine would not satisfy the integration requirements, and integration problems surface late unless deliberately tested early.	The database structure and field names are agreed in writing before any application work begins, and one end-to-end path is built early.
3	R04 — Threshold values prove unsuitable	6	Alerts are the point of the system. Thresholds that fire when nothing is wrong, or stay silent when something is, would undermine the prototype in front of the stakeholder.	Every value is configurable per location and records its basis. The stakeholder confirmed on 3 September 2026 that the ranges may be refined once the platform is established.

14. Risk Monitoring

The register is reviewed at each weekly progress meeting. For each open risk the delivery team records whether the likelihood or impact has changed, whether the mitigation has been carried out, and whether the risk can be closed. New risks identified during design or development are added with the same fields.

A risk that materialises is recorded as an issue in the Challenges and Actions sheet of the project progress tracker, together with the action taken, and its status in this register changes to Occurred.

Status	Meaning
Open	Identified, mitigation in place, not yet materialised
Monitoring	Warning signs present, under active observation
Occurred	The risk has materialised and is now an issue
Closed	No longer able to affect the project

All twelve risks are Open at the date of this document. None has materialised during Phase 1.